Knowledge-Guided Brain Tumor Segmentation

Synchronized visual-semantic-topological prior fusion (STPF)
Zhang & Pan - *BMC Medical Imaging* (2026)

Core contribution. STPF augments conventional MRI feature learning with three explicit knowledge priors: pathology-driven inter-sequence visual differences, machine-generated anatomical/semantic descriptors, and geometric/topological constraints derived from persistent homology. These priors are fused adaptively at both voxel and sample levels.

1. Motivation

The paper argues that strong segmentation networks remain predominantly data-driven and may struggle with ambiguous boundaries, anatomically implausible structures, and the nested relationship among enhancing tumor (ET), tumor core (TC), and whole tumor (WT). STPF therefore attempts to inject radiological semantics and geometric constraints directly into the segmentation process.

2. Three knowledge priors

Prior	Construction	Intended role
Visual/pathological	T1ce-T1, T2-FLAIR and T1/T2 differential channels	Make enhancement, edema and necrotic contrast explicit
Semantic/anatomical	Unsupervised anomaly regions -> standardized attributes/descriptions -> spatialized semantic tokens	Provide location, morphology and signal cues without manual text labels
Topological/geometric	Persistent homology -> skeleton graphs -> GAT tokens -> voxel-space spatialization	Suppress isolated/implausible structures and preserve meaningful connectivi

The original four MRI channels are expanded to a seven-channel enhanced representation. A dual-branch encoder separately processes the original sequences and differential channels, then combines them through cross-attention.

3. STPF architecture

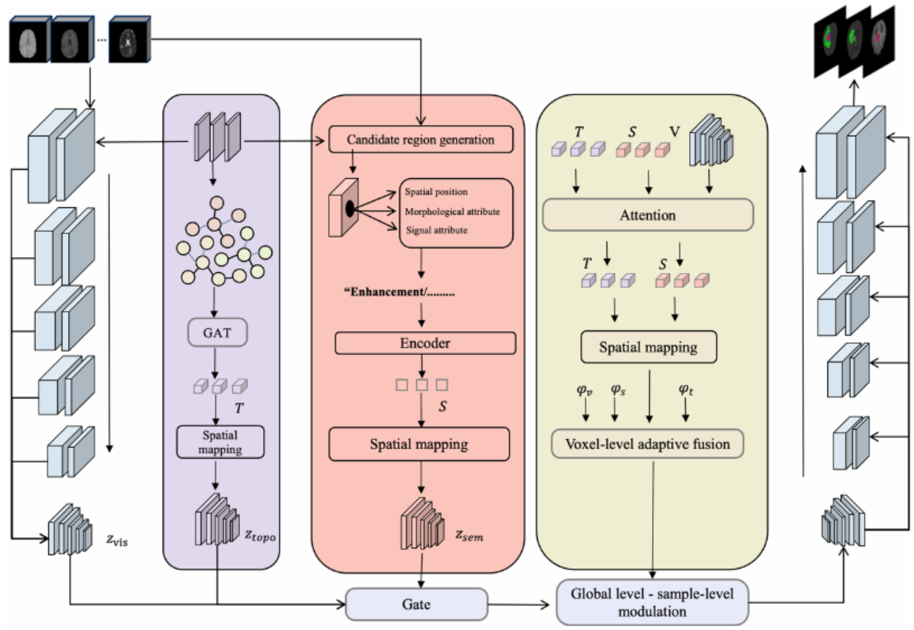


Fig. 1 STPF framework diagram. The left encoder extracts visual features from multi-sequence MRI, obtaining z_vis at the bottom. The purple topological path generates topological tokens T via GAT and obtains z_topo through spatialization. The pink semantic path includes candidate region generation and attribute extraction, obtaining semantic tokens S through encoder and z_sem through spatialization. Three-way knowledge priors are fused at the bottom through gate and sent to global level–sample-level modulation. The yellow-green section performs attention on T, S, V, then completes voxel-level fusion through spatialization and voxel-level adaptive fusion (estimating weights via ϕ_v, ϕ_s, ϕ_t), progressively restoring on the right side and outputting segmentation results

Article Figure 1 (p. 5). The full STPF pipeline. The visual encoder is accompanied by a topological path using GAT-based graph tokens and a semantic path producing spatialized semantic tokens. The decoder performs global sample-level modulation and local voxel-level adaptive fusion.

The fusion mechanism first creates a sample-level conditional vector that drives a hypernetwork and modulates decoder features globally. At each voxel, confidence estimates determine how strongly the model should rely on visual, semantic, or topological information. A unified logit-space fusion stage then coordinates the three sources.

Nested output heads structurally enforce **ET subset TC subset WT**, rather than relying only on a soft penalty to encourage the clinically required hierarchy.

4. Dataset and experimental design

Evaluation uses **BraTS 2020**: 369 multimodal MRI scans with T1, T1ce, T2 and FLAIR, registered to a common template at 1-mm isotropic resolution. The authors use five-fold cross-validation, with approximately 221 training, 74 validation and 74 test cases per fold. Preprocessing includes Z-score normalization, 128 x 128 x 128 spatial crops, differential-channel construction and extensive geometric/intensity augmentation.

Training uses PyTorch on an NVIDIA A100 40GB GPU, AdamW at 2e-4, batch size 2, 300 epochs and cosine annealing. Evaluation uses Dice similarity coefficient (DSC) and HD95; inference uses eight-fold test-time augmentation plus removal of small isolated components and morphological closing.

Table 2 Performance comparison on BraTS 2020 dataset. Reports Dice coefficient (DSC, ↑) and 95% Hausdorff distance (HD95, ↓, unit: mm) for three sub-regions. Best results in bold, second-best underlined.

Method	WT		TC		ET		Mean DSC
	DSC	HD95	DSC	HD95	DSC	HD95	
Attention-UNet [55]	0.845	15.174	0.782	16.380	0.716	9.095	0.781
DAUnet [56]	0.898	5.400	0.830	9.800	<u>0.786</u>	27.600	0.838
SegResNet [29]	<u>0.915</u>	3.275	0.836	3.769	0.730	3.486	0.827
nnU-Net [12]	0.912	3.781	<u>0.842</u>	7.771	0.765	18.230	0.840
TransBTS [32]	0.911	3.360	0.836	<u>2.986</u>	0.740	3.403	0.829
TransUNet [33]	0.892	3.146	0.825	**2.891**	0.758	3.621	0.825
UNETR [31]	0.902	4.305	0.813	5.740	0.732	4.643	0.816
Swin UNETR [13]	**0.917**	2.856	0.826	4.314	0.749	4.503	0.830
SwinBTS [34]	0.891	8.560	0.804	15.780	0.774	26.840	0.823
CKD-TransBTS [35]	0.898	<u>2.419</u>	0.841	3.447	0.770	<u>3.018</u>	0.836
FDiff-Fusion [17]	0.905	**2.207**	<u>0.844</u>	3.311	0.776	**2.714**	<u>0.842</u>
STPF (Ours)	0.901	4.203	**0.864**	4.498	**0.838**	3.217	**0.868**

Table 3 Five-fold cross-validation results (Dice coefficients). Standard deviations in table are cross-fold standard deviations ($n = 5$), used to evaluate model stability under different data splits

	Complete Model (STPF)			MRI-Only Visual Baseline		
Fold	**WT**	**TC**	**ET**	**WT**	**TC**	**ET**
Fold-1	0.904	0.867	0.842	0.866	0.804	0.754
Fold-2	0.901	0.863	0.837	0.860	0.796	0.745
Fold-3	0.898	0.862	0.836	0.859	0.795	0.743
Fold-4	0.903	0.866	0.839	0.865	0.802	0.751
Fold-5	0.899	0.864	0.838	0.861	0.798	0.747
Mean	**0.901**	**0.864**	**0.838**	**0.862**	**0.799**	**0.748**
SD	0.003	0.002	0.002	0.003	0.004	0.004
CV	0.33%	0.23%	0.24%	0.35%	0.50%	0.54%
95% CI	[0.898, 0.904]	[0.862, 0.867]	[0.836, 0.841]	[0.858, 0.866]	[0.794, 0.804]	[0.742, 0.754]

Table 4 Detailed distribution statistics of Dice coefficients ($n = 369$). Based on aggregated out-of-fold prediction results from five-fold cross-validation

Statistical Metric	WT	TC	ET

FLAIR signals of large-scale edema provide sufficient discriminability for pure visual methods. Nevertheless, Table 3 shows STPF's cross-case standard deviation is smaller than pure visual baselines, thereby reflecting bet-

Article Table 2 (p. 11). STPF is compared with attention U-Nets, SegResNet, nnU-Net, Transformer-based models, Swin variants and hybrid architectures under a common BraTS 2020 evaluation

Main result: mean Dice = **0.868**, 2.6 percentage points above the strongest reported baseline in the study. STPF reports TC Dice 0.864 and ET Dice 0.838; ET benefits particularly strongly from semantic/topological correction.

5. Robustness and ablation evidence

Five-fold cross-validation produces coefficients of variation of only 0.23%-0.33%, indicating low fold-to-fold variability. The paper reports narrow confidence intervals and statistically significant improvements over the MRI-only baseline across folds.

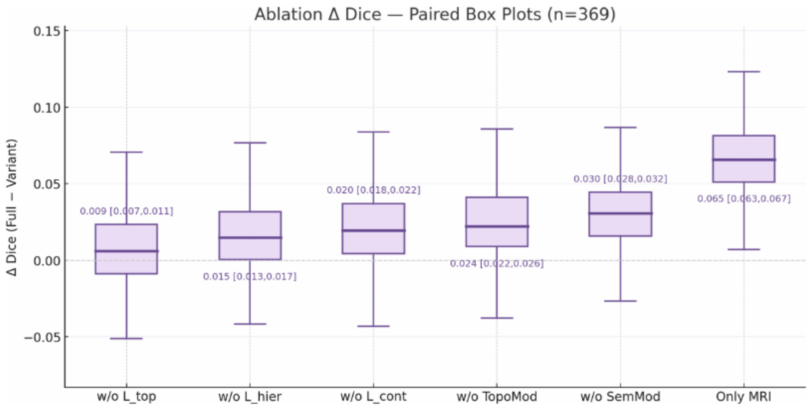


Fig. 4 Ablation experiment Dice coefficient change boxplots ($n = 369$). Each configuration shows performance degradation relative to complete model (ΔDice) and its 95% confidence interval. Removal of semantic prior (w/o SemPrior) and pure visual baseline (only MRI) cause substantial performance loss, thereby proving the important role of knowledge-guided multi-modal prior fusion

Table 7 Paired sample t-tests for ablation experiments ($n = 369$).

Comparison	t-statistic	p-value	95% CI	Cohen's d
w/o $\mathcal{L}_{\mathrm{topology}}$ vs Complete	-7.362	< 0.001	$[-0.011, -0.007]$	0.383
w/o $\mathcal{L}_{\mathrm{hierarchy}}$ vs Complete	-12.269	< 0.001	$[-0.017, -0.013]$	0.639
w/o $\mathcal{L}_{\mathrm{continuity}}$ vs Complete	-16.360	< 0.001	$[-0.022, -0.018]$	0.851
w/o Topological prior vs Complete	-19.640	< 0.001	$[-0.026, -0.022]$	1.022
w/o Semantic prior vs Complete	-24.537	< 0.001	$[-0.032, -0.028]$	1.278
MRI-only visual vs Complete	-53.172	< 0.001	$[-0.067, -0.063]$	2.768

Robustness analysis

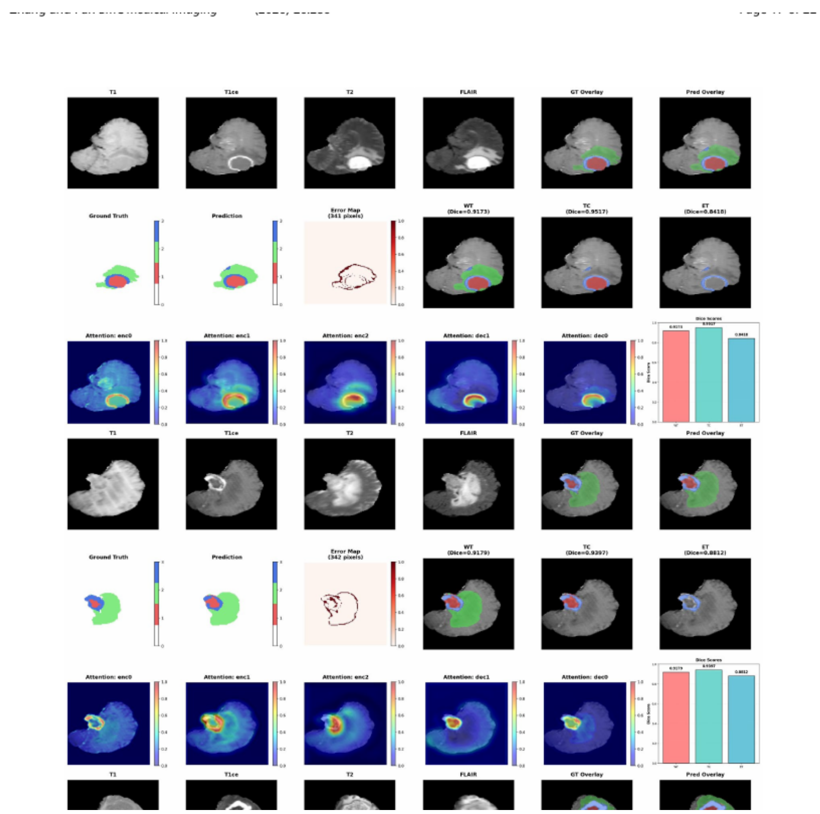
To evaluate model tolerance to incomplete or noisy semantic priors, we designed gradient perturbation experiments covering from mild perturbations to completely random (Table 8).

Perturbation experiments verify adaptive tolerance capability of dual-level fusion mechanism (Fig. 5). Under attribute loss scenarios, performance shows gradual degradation curve (30% loss –0.46%, 50% –0.92%, 70% –1.61%). This is attributed to local confidence weighting detecting semantic incompleteness and automatically elevating visual and topological weights for compensation. Noise injection scenarios show model suppression capability for erroneous information (30% noise only –0.81%), because confidence estimation detects inconsistencies through cross-modal attention and reduces

Article ablation visualization (p. 15). Removing knowledge-guided components degrades Dice. The paper reports particularly substantial losses when semantic and topological priors are removed, supporting the claim that they complement the visual stream rather than merely adding parameters.

The abstract reports a 3.5% degradation after removing semantic priors and a 2.8% degradation after removing topological priors. The MRI stream still supplies most predictive information, but the additional priors are most valuable where visual evidence is ambiguous.

6. Qualitative interpretation and limitations



Article qualitative results (p. 17). Example cases combine multimodal MRI, ground truth/prediction overlays, subregion results, error maps, attention heatmaps and Dice comparisons to illustrate behavior on large and ambiguous-boundary gliomas.

The method assumes all four standard BraTS MRI sequences are available at training and inference. Semantic descriptions are machine-generated weak descriptors rather than neuroradiologist-authored reports, and the authors explicitly identify formal reader validation as future work. The study is also benchmark-centered; broader external clinical validation remains necessary.

7. Overall conclusion

STPF is a notable example of **knowledge-guided medical AI**. Instead of asking a network to discover every useful relationship from image intensities alone, it explicitly supplies radiological contrast patterns, coarse anatomical semantics and topological structure, then learns when and where each source should influence the prediction. The reported BraTS 2020 gains suggest that this strategy is especially useful for difficult boundaries and small enhancing regions.

Future directions proposed by the authors include semi-/few-/weakly-supervised settings, heterogeneous modalities and medical foundation models, and evaluation on newer BraTS tasks with architectures such as SegMamba and Diff-UNet.